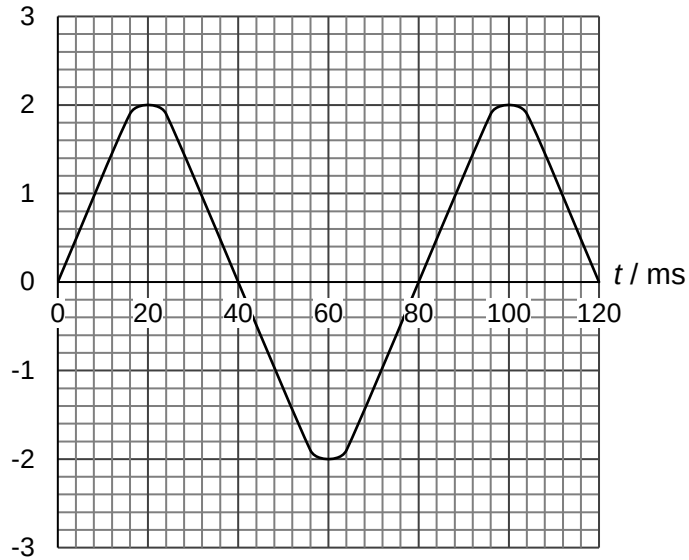
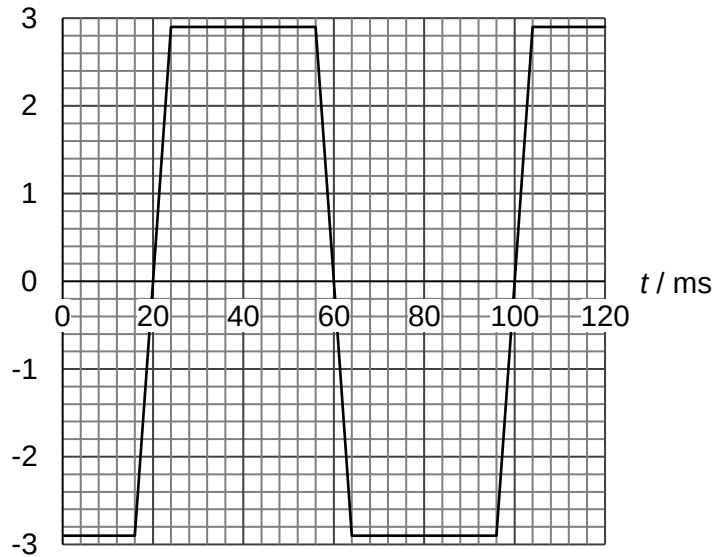


	$\text{work done} = (-7.8 \times 10^{-5}) - (-2.0 \times 10^{-5}) = 1.0 \text{ J}$	
4(a)	The magnetic flux linkage is the product of the number of turns of the coil of wire and the magnetic flux through (each turn of) the coil of wire. The magnetic flux through an area is defined as the product of that area and the component of the magnetic flux density normal to the plane of that area.	[1] [1]
4(b)(i)	$\begin{aligned}\phi_{\max} &= B_{\max} A \\ &= (\mu n I_{\max}) A \\ &= (1000 \times 4\pi \times 10^{-7}) \left(\frac{1800}{0.12} \right) (1.05 \times 10^{-3}) \left(\frac{\pi \times 0.036^2}{4} \right) \\ &= 2.014 \times 10^{-5} \text{ Wb} \\ &= 2.0 \times 10^{-5} \text{ Wb}\end{aligned}$	[1] substitution for $B_{\max}$ only [1] unrounded value for max flux
4(b)(ii)	By Faraday's law of EMI, $\begin{aligned}E_{\text{induced}} &= -\frac{d\Phi}{dt} \\ &= -\frac{d(N_Q BA)}{dt} \\ &= -N_Q A \frac{d(1000\mu_0 n I)}{dt} \\ &= -1000 N_Q A \mu_0 n \frac{dI}{dt}\end{aligned}$ $\frac{dI}{dt} = \frac{[1.0 - (-1.0)] \times 10^{-3}}{(24 - 56) \times 10^{-3}}$ $\begin{aligned}E_{\text{induced}} &= -(1000)(2400) \left(\frac{\pi(0.036)^2}{4} \right) (4\pi \times 10^{-7}) \left(\frac{1800}{0.12} \right) \frac{dI}{dt} \\ &= 2.88 \text{ V}\end{aligned}$	[1] for correct substitution of $\mu n A$ [1] correct gradient from Fig. 4.2 [1] for correct substitution of N_Q [1] for final answer

4(b)(iii)

 $\phi / \times 10^{-5} \text{ Wb}$  V / V [1] for $\phi - t$ graph[1] for $V - t$ graph

[1] for correct scale on vertical axes of both graphs